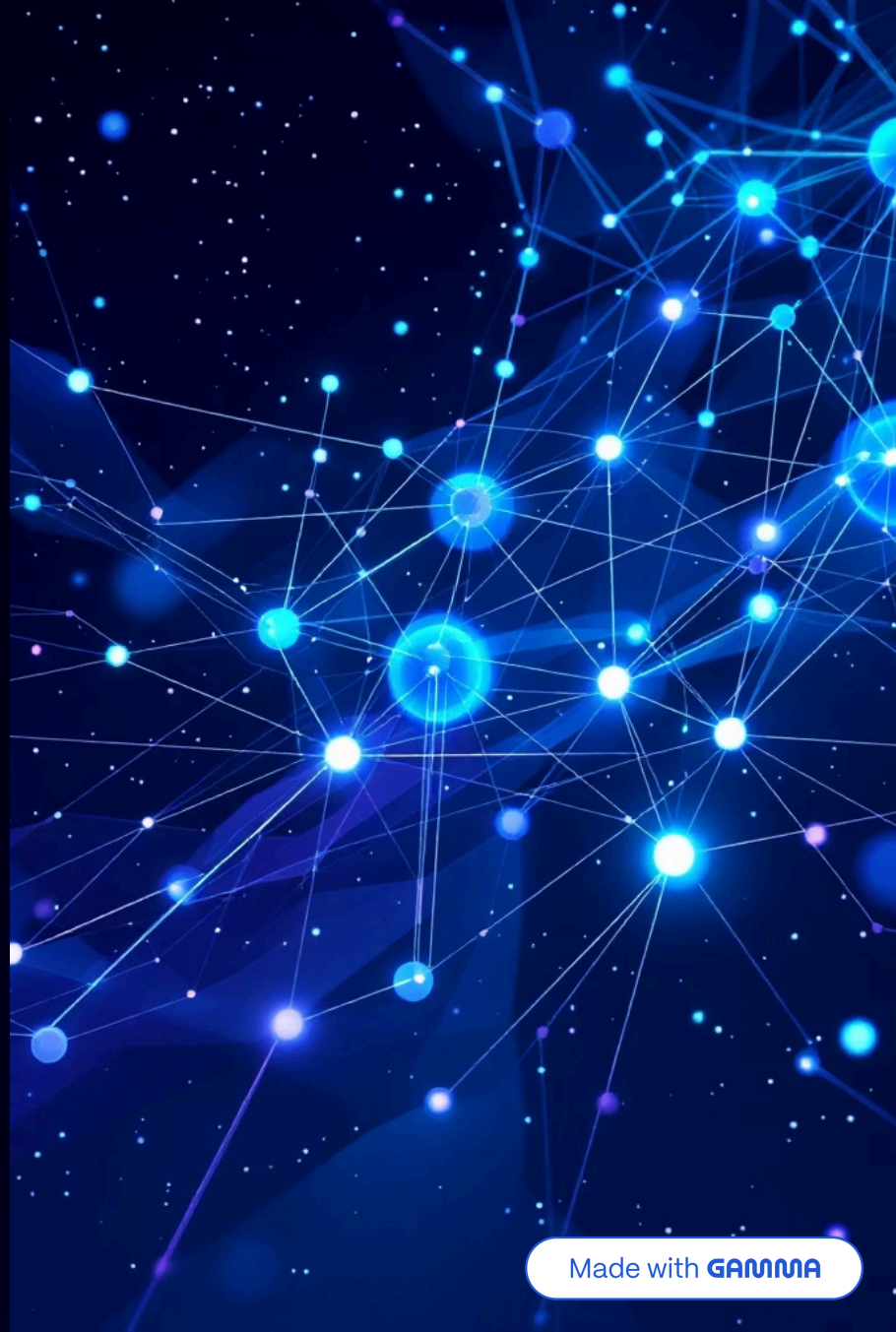
TrustLens v1

AI Reliability Evaluation System

Measuring agreement to estimate trust in AI responses through semantic similarity analysis



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The Challenge: AI Trust Gap

Current AI Limitations

Modern language models hallucinate facts, generate conflicting answers to identical queries, and project unwavering confidence regardless of accuracy. Users face an impossible question:

> "Is this answer reliable?"

The market demands a trust estimation layer—not better AI, but AI evaluation infrastructure.

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The TrustLens Solution

Semantic Similarity

Transforms multiple AI responses into vector embeddings using state-of-the-art language models

Consensus Score

Computes agreement metrics through cosine similarity calculations across high-dimensional vector space

Reliability Signal

Outputs interpretable trust scores that users can integrate into decision-making workflows

Core insight: We're not building better AI—we're building the evaluation layer that makes AI trustworthy

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Mathematical Foundation

Everything operates in vector space. Each answer becomes a point in high-dimensional space, where semantic meaning maps to geometric direction.

$$\cos(\theta) = \frac{A \cdot B}{\|A\|\|B\|}$$

High Agreement

Vectors point in similar directions, producing cosine values near 1.0. Multiple AI models converge on identical semantic meaning.

Low Agreement

Vectors diverge significantly, yielding cosine values near 0. AI models generate conflicting interpretations of the same query.

The average similarity across all response pairs produces the **Consensus Score**—a single reliability metric.

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System Architecture

User Query

Single question submitted to evaluation system

Embeddings Model

SentenceTransformer encodes responses as vectors

Vector Comparison

Pairwise cosine similarity calculations

Score Aggregation

Consensus score generation

UI Display

Streamlit interface shows reliability metrics

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Demo: Four Response Patterns

1

High Agreement (Correct)

Multiple models converge on factually accurate answer. **Result:** High consensus score (0.85+). System correctly identifies reliable information.

2

Mixed Answers

Some agreement on core concepts but variations in detail. **Result:** Medium score (0.4-0.6). Flags need for human review or additional verification.

3

Conflicting Responses

Models generate contradictory answers to identical query. **Result:** Low score (0.2-0.3). System detects uncertainty and warns user.

4

False Consensus

All models agree but propagate identical factual error. **Result:** High score despite inaccuracy. *Reveals research opportunity for external fact grounding.*

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Real-World Applications

Education AI Validation

Verify accuracy of AI tutoring systems before deployment in classrooms. Ensure students receive reliable guidance on homework and exam preparation.

Healthcare AI Cross-Check

Audit diagnostic reasoning models for consistency before clinical use. Flag unreliable symptom analysis or treatment recommendations.

News Misinformation Filtering

Identify conflicting narratives across AI-generated news summaries. Detect propaganda or biased reporting through consensus analysis.

Multi-LLM Comparison

Objectively evaluate GPT-4, Claude, Gemini, and other models on identical queries. Generate performance benchmarks for enterprise procurement.

Enterprise AI Auditing

Continuous monitoring of production AI systems for reliability drift. Compliance reporting for regulatory requirements and internal governance.

Research Tool

Study hallucination patterns across model architectures and training datasets. Quantify uncertainty in frontier AI capabilities.

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Future Roadmap

01

Multi-Model Voting

Ensemble methods combining diverse AI architectures

02

External Fact Grounding

Database verification to break false consensus

03

Uncertainty Modeling

Probabilistic confidence intervals

04

Weighted Consensus

Model performance-based weighting schemes

05

Confidence Heatmap

Visual analytics for agreement patterns

TrustLens v1 establishes the evaluation layer. Future versions will close the gap between agreement and truth.

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